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Lecture 3

Financial Market

Economics B — Macroeconomics I

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University of St.Gallen

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Learning objectives and literature of the 3rd lecture

- ▶ Learning Objectives:
 - ▶ The role of money in the national economy
 - ▶ Money supply and demand
 - ▶ Monetary policy through central banks
 - ▶ Financial market for a model of the economy in the short run
 - ▶ Derivation of the LM curve for the IS-LM model
- ▶ Literature: Textbook by Blanchard, chapter 4 & 5.2

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- ▶ The financial market plays a central role in an economy. It is complementary to the goods market.
- ▶ Financial market is a generic term for a market in which trading of financial instruments takes place:
 - ▶ *Money market*: Short-term money supply and demand
 - ▶ *Bonds market* (also credit market): financial assets and liabilities
 - ▶ *Capital market*: medium and long-term capital requirements and supply

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Nomenclature (continued)

- ▶ Assumption: Only two forms of assets, money and bonds.
This allows us to understand how the interest rate is set and the role of the central bank in this process.
 - ▶ Money can be used for transactions, but does not earn interest.
 - ▶ Bonds yield an interest rate i , but cannot be used for transactions.
- ▶ Individuals choose what fraction of their wealth they want as money and what fraction as bonds.

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What is money?

- ▶ The functions of money were discussed in Economics A, Microeconomics
 - ▶ medium of exchange, unit of account, means of storing value
- ▶ In lecture 6 we will discuss how inflation affects the functioning of money.
- ▶ Historically, there have been many forms of money:
Stones, shells, gold, receipts, cigarettes, Bitcoin, ...

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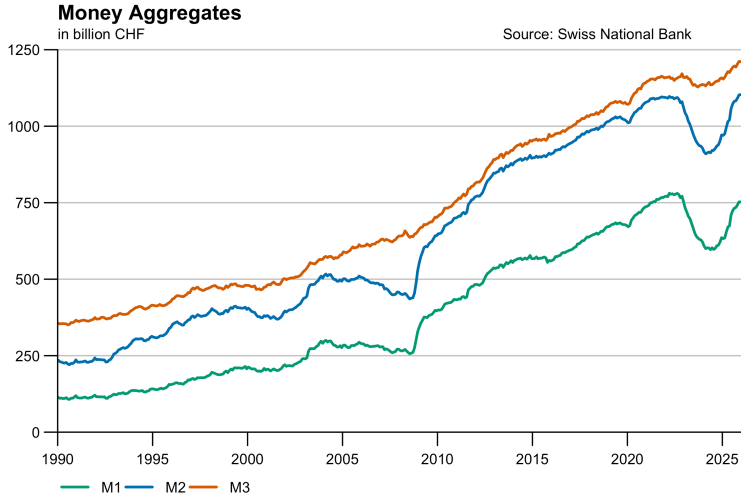
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Types of money in a modern economy

- ▶ Central banks measure money supply using monetary aggregates.
- ▶ Definitions used by the Swiss National Bank (SNB):
 - ▶ Monetary base: sum of banknotes in circulation plus sight deposits of domestic commercial banks held at the SNB
 - ▶ M1: currency plus sight deposits in Swiss francs from residents
 - ▶ M2: M1 plus savings deposits held at banks ($\neq$ pillar 2, 3a) in Swiss Francs
 - ▶ M3: M2 plus time deposits held at banks in Swiss Francs

Types of money in a modern economy II



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Demand for money

- ▶ We describe the money market with supply and demand.
- ▶ Individuals choose whether they hold their assets as money or bonds.
Motives for holding money: transactions, caution, speculation
- ▶ The demand for money: $M^d = P Y * L \left(\begin{smallmatrix} i \\ (+) \quad (-) \end{smallmatrix} \right)$
 - ▶ Increases proportionally with nominal income ($P Y$)
 - ▶ Depends on the interest rate i
→ $L(i)$ is a function of the interest rate.
- ▶ Extensions are possible: M^d might depend on costs of liquidating bonds.

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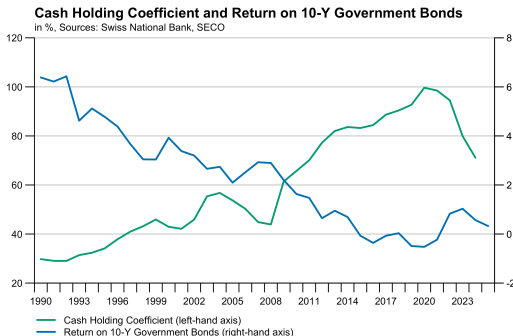
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Demand for money – Look at the data

- ▶ How well does the money demand relation reflect reality?
- ▶ $M^d = P Y * L(i) \longleftrightarrow \frac{M^d}{P Y} = L(i)$
- ▶ Cash-deposit ratio $L(i)$ = money holding / nominal income.
- ▶ Empirical evidence: Switzerland, 1990–2025



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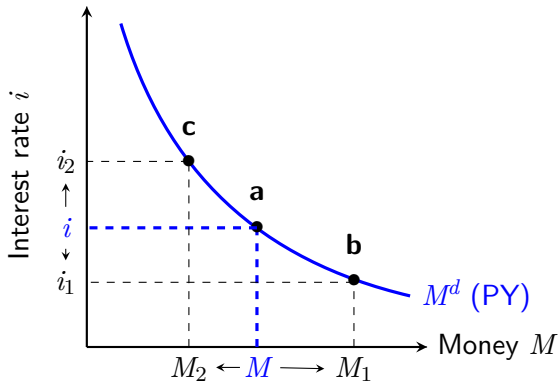
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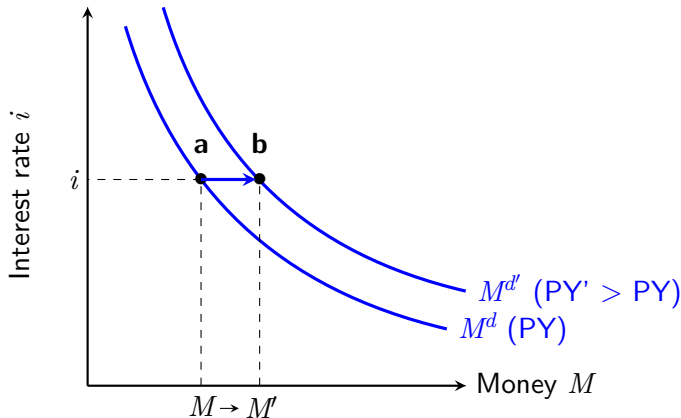
Demand for money – Graph

- For a given nominal income (PY) a lower interest rate increases the demand for money. In contrast, as the interest rate rises, the liquidity preference and thus the demand for money decreases.



Demand for money– Graph (continued)

- At a given interest rate, an increase in nominal income raises the demand for money, leading to a shift to the right of the money demand curve.



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Money supply

- ▶ Money is provided by central banks (cash and reserves for commercial banks) and by commercial banks (demand deposits).
 - ▶ Money creation by central banks and commercial banks (see slide 21)
- ▶ Instruments of monetary policy by central banks:
 - ▶ Open market operations: buying and selling assets (foreign exchange, government bonds, corporate bonds) in the open market (referred to as '*Quantitative Easing* (QE)' for long-term bonds)
 - ▶ Choosing the money supply (late 1970s to early 1990s, revival through QE) vs. choosing the interest rate (since the early 1990s)
 - ▶ Expectation management with regard to future policy: *forward guidance*
 - ▶ Reserve rate and other incentives for commercial banks
 - ▶ Helicopter money

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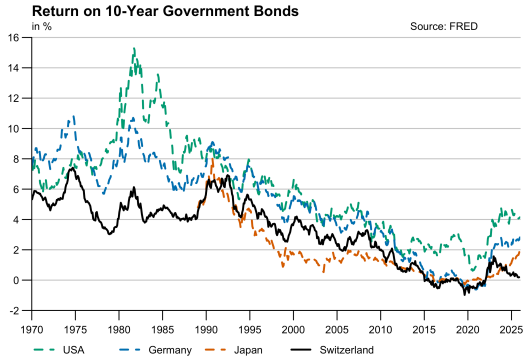
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Money supply — Open market operations

- ▶ In modern economies all central banks conduct monetary policy through open market operations, usually through the purchase and sale of *government bonds*.
- ▶ With an expansionary open market policy, the balance sheet of the central bank expands:
 - ▶ assets: additional bonds
 - ▶ liabilities: additional money supply
- ▶ The opposite of expansionary monetary policy is contractionary monetary policy.
- ▶ Relationship between interest rates and bond price:
 - ▶ There is an inverse relationship between bonds price and interest rate.
 - ▶ Price of a bond with a one-year maturity: $P_B = 100/(1 + i)$.

Interest rates in practice

- ▶ Interest rates fell worldwide and for decades:



- ▶ *Zero lower bound*: how negative can an interest rate be?
 - ▶ Cash pays zero interest, but holding it causes costs.

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Interest rate in equilibrium

- ▶ An equilibrium on the money market is achieved when the supply of money equals the demand for money.

$$\text{▶ } M^s = M^d \quad \longleftrightarrow \quad M^s = P Y * L(i)$$

- ▶ The properties of the equilibrium are contingent on whether the central bank uses the money supply or the interest rate as an instrument.
- ▶ In the first case we speak of money supply control, in the second of interest rate control.

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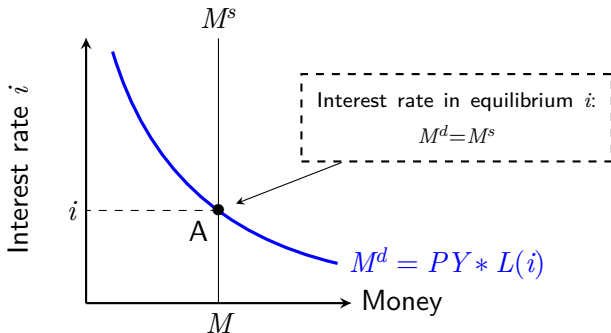
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Money demand, money supply and the equilibrium interest rate

- *Money supply control*: The central bank provides money supply M , so that $M = M_S$.
- The interest rate in equilibrium: the (interest-dependent) demand for money is equal to the supply of money.



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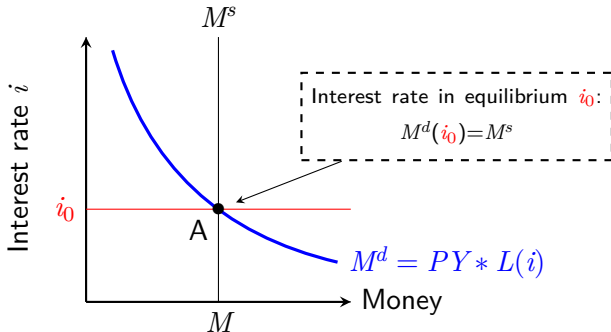
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Money demand, money supply and the equilibrium interest rate

- *Interest rate control*: The central bank sets the interest rate to i_0 .
- The central bank provides as much money at the fixed interest rate i_0 as is requested at that rate: $M^d(i_0)$. The money supply (M^s) is endogenous.



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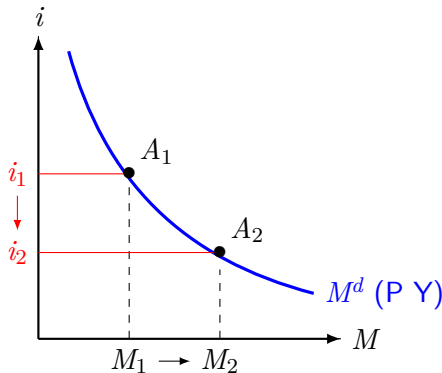
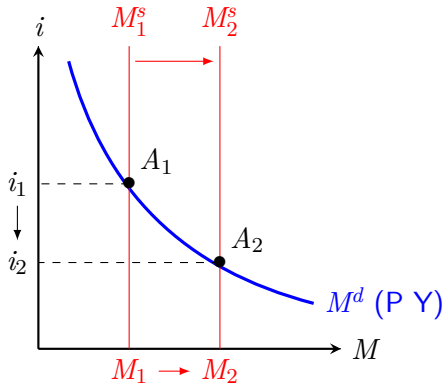
Conclusion

Expansionary monetary policy through open market operations

- ▶ Money supply control:
 - ▶ Increasing the money supply from M_1 to M_2 shifts the money supply curve to the right
 - ▶ Equilibrium interest rate falls from i_1 to i_2 .
 - ▶ The demand for money is rising accordingly.
 - ▶ → The interest rate is determined endogenously.
- ▶ Interest rate control:
 - ▶ The central bank is aiming for a lower interest rate.
 - ▶ If the interest rate falls from i_1 to i_2 , the demand for money increases from M_1 to M_2 .
 - ▶ The central bank increases the money supply to M_2 (point A_2).
 - ▶ → The money supply is determined endogenously.

Expansionary monetary policy through open market operations

- Graphs: Money supply control (left) and interest rate control (right)



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Money creation by commercial banks

- ▶ Private commercial banks as well as central banks determine the amount of money available.
- ▶ In an economy with commercial banks, the central bank can only control the monetary base H directly, the money supply M is controlled indirectly.
- ▶ We distinguish between:
 - ▶ M = money supply
 - ▶ H = central bank money (also *high-powered money*, monetary base or $M0$)
 - ▶ CU = cash in circulation
 - ▶ D = deposits of non-banks

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Money creation by commercial banks (continued)

- ▶ Commercial banks do not lend 100% of deposits, but hold reserves for three reasons:
 - ▶ Deposits and withdrawals of investors are not equal.
The commercial bank must always have a certain amount of cash on hand.
 - ▶ To cover debts to other banks.
 - ▶ In order to fulfill the legal reserve requirements.
- ▶ Commercial banks are lending the majority of deposits.
 θ is the reserve ratio, the amount of reserves banks hold per dollar of deposits. They deposit this reserve with the central bank.
- ▶ The reserve is: $R = \theta * D$

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Money creation by commercial banks — Money creation multiplier

- ▶ The monetary base H consists of cash in circulation plus reserve deposits of the banks:

$$\text{▶ } H = CU + R \iff H = CU + \theta D$$

- ▶ Assumption: cash and deposits amount to a fixed proportion (b and $1 - b$) of the total money in circulation M . Then:
- ▶ If the central bank increases the monetary base H by one unit, how does this affect M ?
 - ▶ The commercial bank can lend a fraction $1 - \theta$ of this, of which the fraction $1 - b$ flows back into the banking system as deposits. This process continues indefinitely, yielding the following geometric series:
$$1 + (1 - \theta)(1 - b) + [(1 - \theta)(1 - b)]^2 + \dots = 1 / (1 - (1 - \theta)(1 - b))$$
- ▶ The money creation multiplier results: $M = \frac{1}{b + \theta(1 - b)} H$

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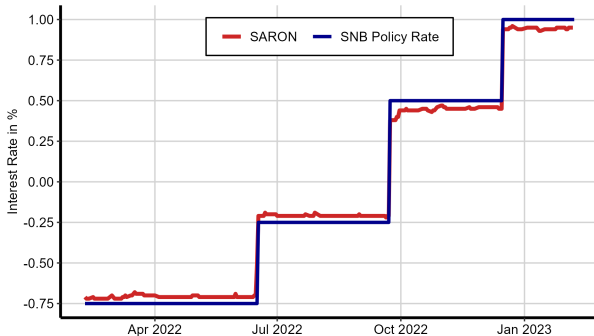
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Money creation by commercial banks — Implications

- ▶ The central bank can now control money supply through three channels:
 - ▶ policy rate (0.00% in Jan'2026): Price at which commercial banks can obtain central bank money from the central bank
 - ▶ deposit rate (0.00% in Jan'2026): Interest on deposits by commercial banks with the central bank
 - ▶ minimum reserve rate (4.00% in Jan'2026): compulsory balances of commercial banks with the central bank
- ▶ The central bank indirectly influences the volume of lending by commercial banks and thus the amount of their deposits and the money supply M .

Monetary policy in practice

- ▶ Example of monetary policy of the SNB: see *NZZ* article, Jan'2023.
- ▶ The SNB had to intervene directly in the market in October 2022 as the SARON differed from the SNB policy rate:



Source: SNB (https://www.snb.ch/de/i/about/stat/statrep/id/current_interest_exchange_rates)

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IS-LM: A Model of the economy in the short-run

- ▶ Repetition: The goal is a (simple) economic model for the analysis of the economy in the short-run.
- ▶ In the last lecture: Balance on the goods market
 - ▶ Implicit IS curve: $Y = c_0 + c_1 * (Y - T) + I(Y, i) + G$
- ▶ On the basis of today's lecture we can add i as an equilibrium result from the financial market.
- ▶ IS-LM model in the diagram with i and Y on the axes.
- ▶ Equilibrium in the financial market in the diagram: the LM curve.
- ▶ “LM” stands for liquidity and money supply.

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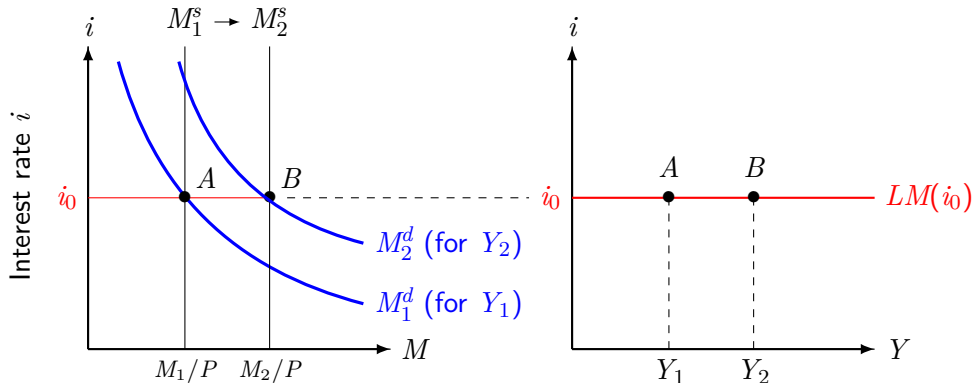
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Derivation of the LM curve under interest rate control

- ▶ The money market is in balance if $M = P Y^* L(i)$
 - ▶ Divide both sides by the price level: $M/P = Y^* L(i)$
 - ▶ The real supply of money is equal to the real demand for money. The latter depends on real income and the interest rate.
 - ▶ Note: in the short term the price level P is fixed.
- ▶ If the money market is in equilibrium, the bonds market will be as well (→ Walras' law).
- ▶ Interest rate control: Central bank sets interest rate i and endogenously adjusts the supply of money to the demand for money at that rate. The LM equation can thus be described by a horizontal line.

Derivation of LM curve for interest rate control

- With higher income $Y_2 > Y_1$ the demand for money is higher:



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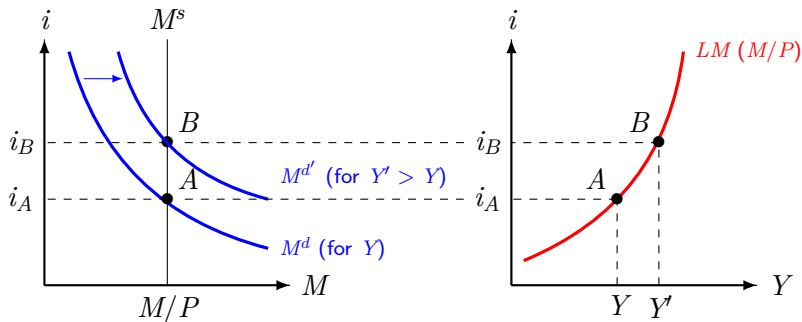
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Derivation of the LM curve in money supply control

- ▶ Money supply control:
 - ▶ Central bank keeps money supply constant as income increases, the interest rate increases from i_A to i_B (see point B).
 - ▶ With a constant supply of money, the relationship between income and interest rate is an LM curve that increases with Y .



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The LM Curve

- ▶ The textbook (usually) assumes that the central bank controls interest rates, and therefore uses a flat LM curve.
- ▶ Summary:
 - ▶ above the LM curve: excess supply on the money market
 - ▶ below the LM curve: excess demand on the money market
 - ▶ on the LM curve: equilibrium on the money market
- ▶ The bonds market here is the mirror image of the money market. The LM curve also describes the balance on the bonds market.

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Summary and outlook

- ▶ We have analyzed the financial market of an economy.
- ▶ The LM curve describes the equilibrium on the financial market in the interest/income diagram.
- ▶ We will combine IS and LM curve in the next lecture:
IS-LM model for the analysis of the economy in the short term.
- ▶ Preparation for lecture 4: Slides and chapter 5 & 6 in the textbook

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Discussion and Quiz

- ▶ Open questions from the auditorium?
- ▶ To practice, we provide some old exam questions in the lecture. Please participate via QR code!